

AI Agents and Workflows for Developers – April 2026

Individual Project Assignment

Overview

You will create a Python Jupyter notebook (.ipynb file) in Google Colab. Your goal is to design and implement a Multi-Agent workflow using **LangGraph** and **LangChain**. The system must solve a complex, multi-step problem of your choosing by utilizing distinct AI agents, custom tools, conversation memory, and a human-in-the-loop review process.

Choose a Scenario

Choose a domain and use case of your interest. Examples include, but are not limited to:

- **Research Assistant:** One agent searches the web, another summarizes the findings.
- **Travel Planner:** One agent checks flights/hotels, another builds the itinerary.
- **Code Reviewer:** One agent writes code based on a prompt, another reviews it for bugs. Define the scenario clearly in a markdown cell at the beginning of your notebook.

Multi-Agent Architecture (LangGraph)

You must build a stateful graph using LangGraph. The system must include:

- At least **two distinct agents**, each with a specific role, system prompt, and responsibility.
- A defined state (TypedDict or Pydantic model) that is passed between the agents and stores the progress of the workflow.

Agents & Tools

Your agents must have access to and successfully use at least **two different tools**. These can be built-in LangChain tools (e.g., Wikipedia search, Tavily web search, Calculator) or custom Python functions wrapped as tools (e.g., a mock API call, a string manipulator).

Memory & Human-in-the-Loop

The graph must maintain conversational memory (e.g., using MemorySaver or checkpointing in LangGraph). Furthermore, you must implement a **Human-in-the-Loop (HITL)** interruption. Before the system outputs its final result or takes a critical action (like "sending an email" or "finalizing a report"), the graph must pause, ask for human approval or feedback, and then resume execution based on the human's input.

Core Function

You need to implement a core function `execute_workflow` that accepts one string parameter – `user_request`. This function should initialize the graph and begin the execution. You must also implement the logic to handle the graph interruption, receive the human feedback, and resume the graph to get the final output.

Test Cases

Write at least **5 test cases** (different initial user requests) that execute your workflow. Show how the agents interact and clearly demonstrate the Human-in-the-Loop process (e.g., simulate the user approving the action in one test case, and providing feedback to revise the action in another).

Security

All your API keys (OpenAI, Anthropic, Tavily, etc.) must be stored in environment variables or Colab Secrets. **DO NOT SUBMIT YOUR API KEYS IN THE CODE!**

Working System

When API keys are set up, the system should be able to execute in the Google Colab environment without any additional installations beyond the required pip install commands at the top of the notebook.

Submission Format

ZIP file containing:

- .ipynb Jupyter Notebook
- An optional README.md or small data file if your specific scenario requires it.